

FROM PROMPT TO GRAPH

five layers we've wrapped around the model, and why each one had to exist

What we're covering

01 PROMPT *when the prompt was the whole program*

02 CONTEXT *what does the model actually get to see?*

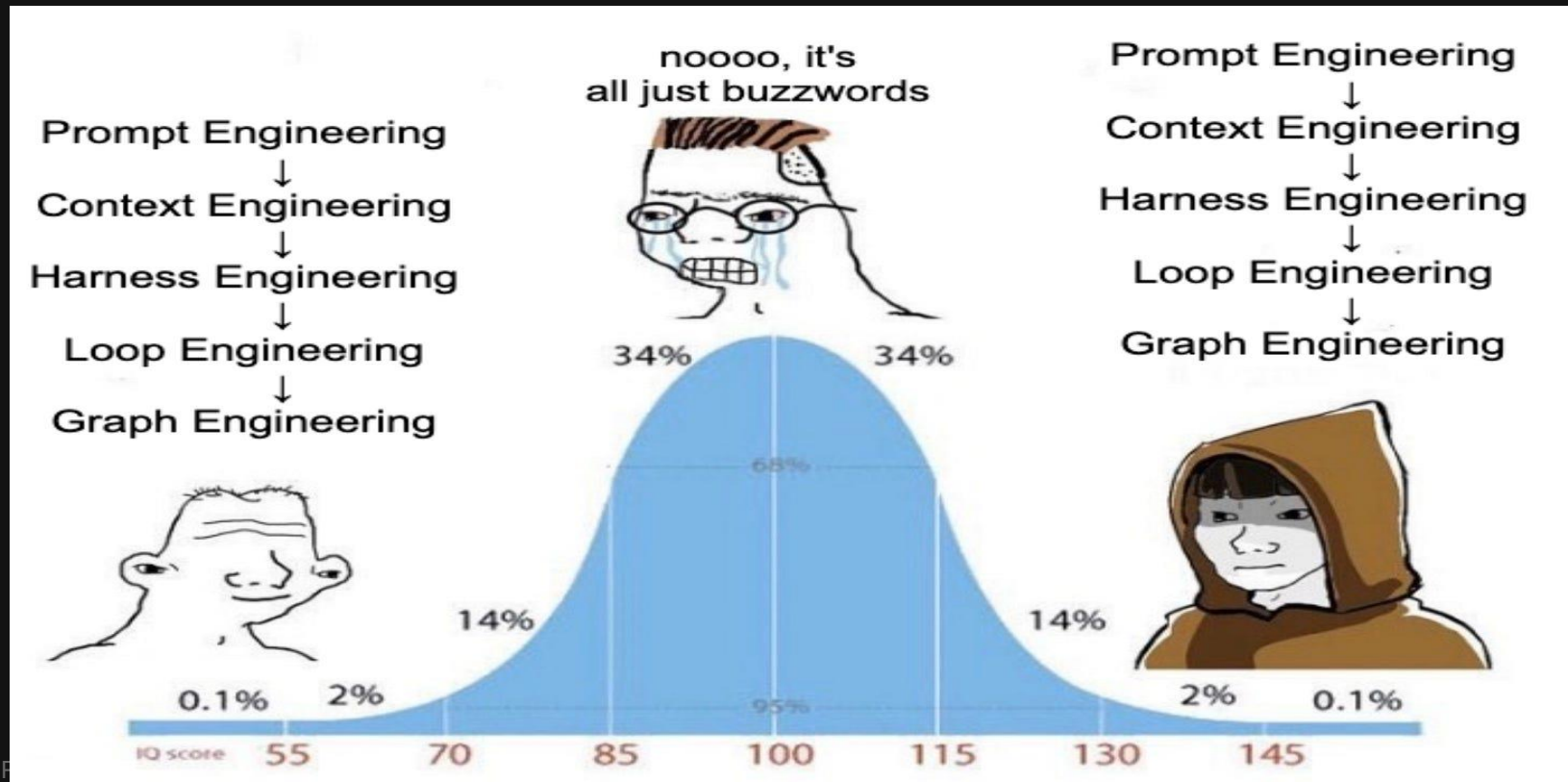
03 HARNESS *what surrounds the model?*

04 LOOP *what happens after the first answer?*

05 GRAPH *or is a graph just a loop that branches?*

Each one wraps the last. None of them replaced it.

KNOWING AND UNDERSTANDING MATTERS!!!!



01

PROMPT

when the prompt was the whole program

THE DEFINITION OF PROMPT ENGINEERING

The process of crafting and refining text inputs to guide ai models towards the human defined goal

PROMPT EXAMPLE

```
prompt_engg.py

from anthropic import Anthropic

client = Anthropic()

FEW_SHOT_PROMPT = """You are a triage classifier for an Apple device support line.
Classify each message into exactly one category: battery, screen_repair, billing,
software_bug, other.
Respond with only the category name.

Message: "My iPhone battery drains to 20% by noon even when I'm barely using it."
Category: battery

Message: "I was charged twice for AppleCare+ this month."
Category: billing

Message: "{ticket}"
Category: ""

def classify_ticket(ticket_text: str) -> str:
    response = client.messages.create(
        model="claude-sonnet-4-6",
        max_tokens=10,
        messages=[{"role": "user", "content":
FEW_SHOT_PROMPT.format(ticket=ticket_text)}],
    )
    return response.content[0].text.strip()

print(classify_ticket("My screen has a hairline crack after a small drop."))
# -> screen_repair
```

2023

NO GRADIENTS. JUST EXAMPLES.

GPT-3 — "Language Models are Few-Shot Learners." The model learns from what's in the prompt, and nothing is trained.

3.2×

Chain-of-thought prompting tripled PaLM 540B's accuracy on GSM8K.

17.9% → 56.9%, from wording alone. Wei et al., Google Research, 2022.

Training language models to follow instructions with human feedback

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OpenAI

A 100× SMALLER MODEL WON.

InstructGPT: labelers preferred the 1.3B tuned model over the 175B base. Following intent beat raw scale.

THE DEFINITION OF RHLF

RHLF stands for Reinforcement Learning with Human Feedback which is technique to train AI models to produce answers that humans prefer

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

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Abstract

General reasoning represents a long-standing and formidable challenge in artificial intelligence. Recent breakthroughs, exemplified by large language models (LLMs) (Brown et al., 2020; OpenAI, 2023) and chain-of-thought prompting (Wei et al., 2022b), have achieved considerable success on foundational reasoning tasks. However, this success is heavily contingent upon extensive human-annotated demonstrations, and models’ capabilities are still insufficient for more complex problems. Here we show that the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL), obviating the need for human-labeled reasoning trajectories. The proposed RL framework facilitates the emergent development of advanced reasoning patterns, such as self-reflection, verification, and dynamic strategy adaptation. Consequently, the trained model achieves superior performance on verifiable tasks such as mathematics, coding competitions, and STEM fields, surpassing its counterparts trained via conventional supervised learning on human demonstrations. Moreover, the emergent reasoning patterns exhibited by these large-scale models can be systematically harnessed to guide and enhance the reasoning capabilities of smaller models.

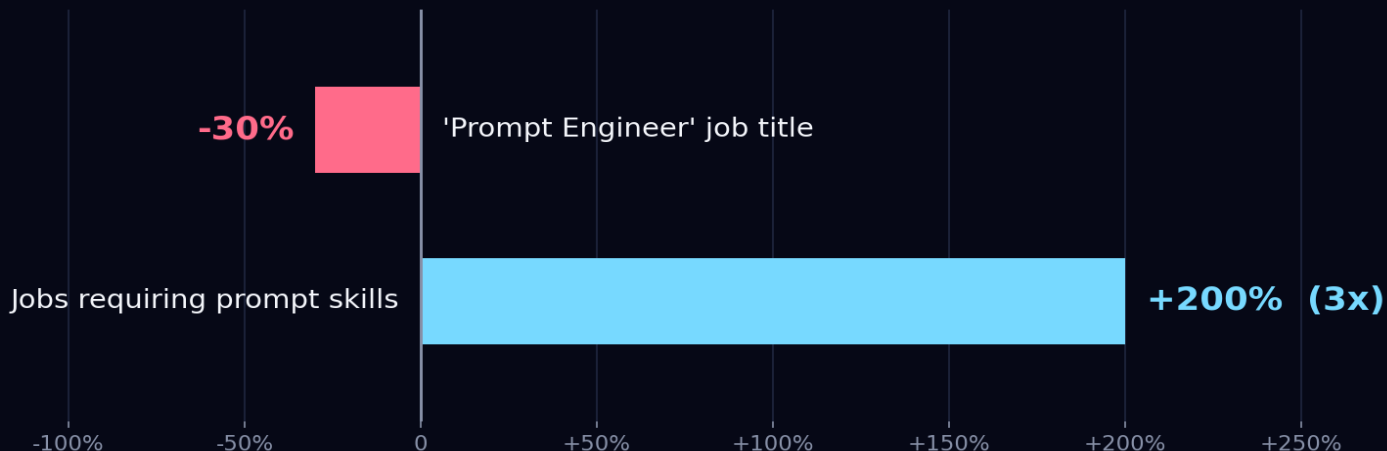
THE DEFINITION OF GRPO

Instead of learning how good a response is with a separate value network, it judges each response relative to other responses generated for the same prompt

2 0 2 4

The job title is dying. The skill is booming.

Change from 2024 to 2026 — the standalone "Prompt Engineer" title vs. jobs that require prompting skills.



The market is hiring the skill, not the title. · Source: LinkedIn 2026 hiring data · [OC]

74.4%

Trained to reason, o1 scored 74.4% on AIME 2024. GPT-4o, prompted, scored 9.3%.

pass@1 at maximal test-time compute. OpenAI, September 2024.

THE REVERSAL

STOP TELLING IT TO THINK.

OpenAI now advises against chain-of-thought prompts. The 2022 trick became a training objective.

But it didn't die. It moved.

STILL BRITTLE

Reword the question, move the score 12%.

*One perturbation flips model rankings in 63% of cases.
Brittlebench, 2026.*

STILL ADVISED

Examples are still "strongly advised."

Anthropic on context engineering, September 2025.

What got absorbed was hand-written chain-of-thought. The rest changed name.

what a prompt still can't be

static. blind. alone.

However well it reads you, it knows nothing you didn't tell it — and there is a hard limit on how much you can tell it.

02

CONTEXT

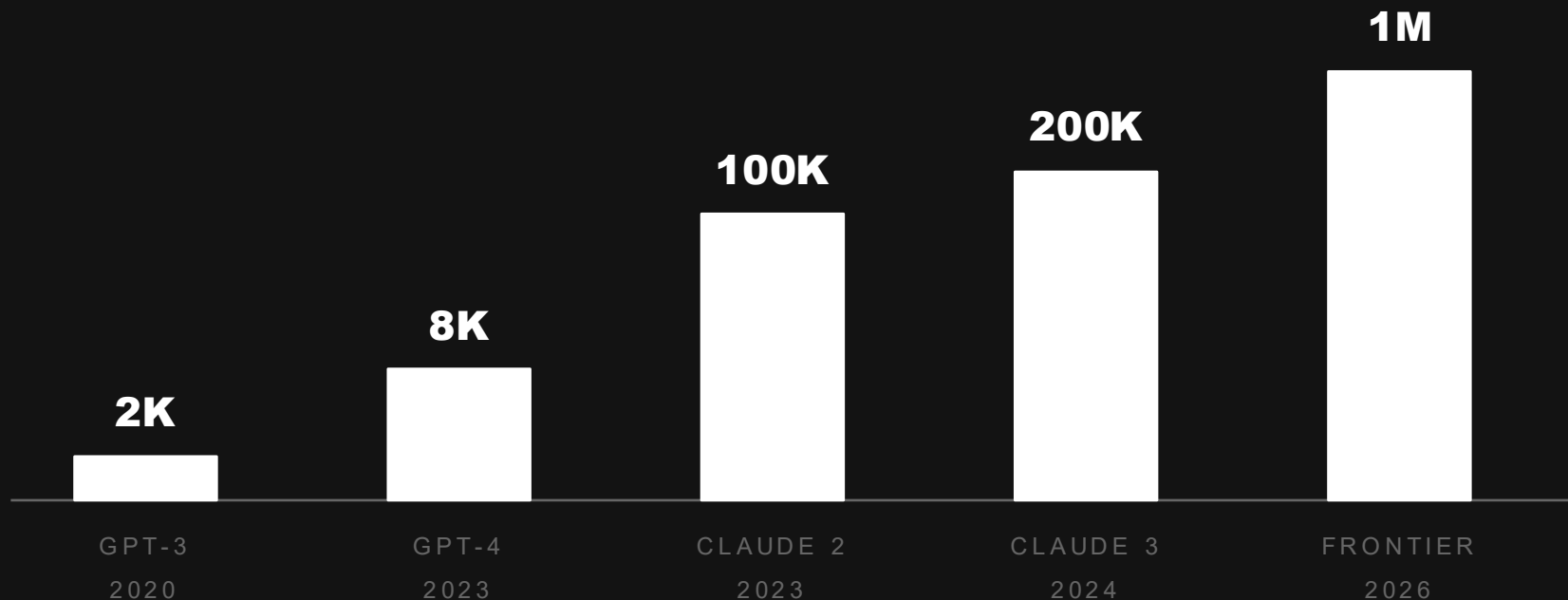
what does the model actually get to see?

THE DEFINITION OF CONTEXT ENGINEERING

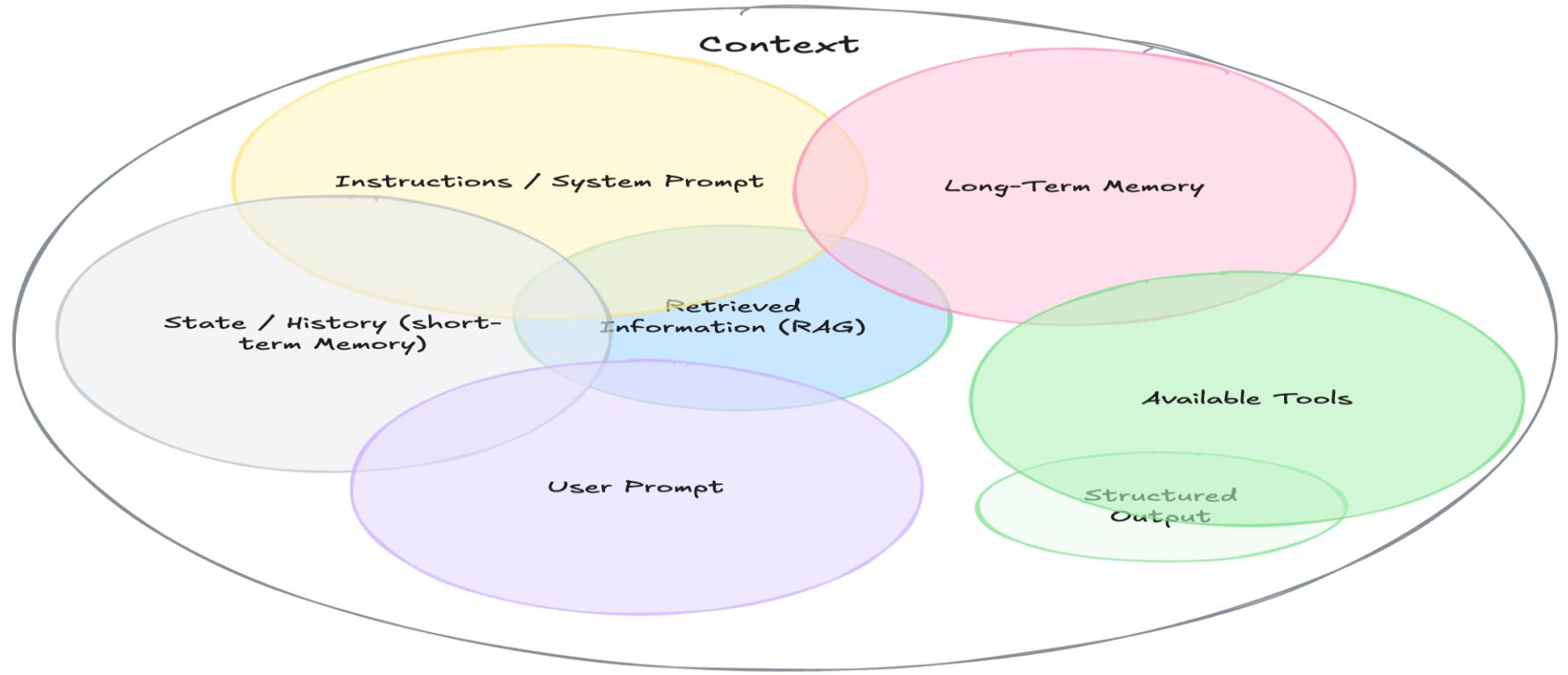
The process of managing the information the model sees before it generates its response

The context window grew roughly 500× in six years.

Tokens, log scale. 2,048 → ~1,000,000.



Context Engineering



instructions & the ask(System & User Prompt)

WHO IT IS, THEN WHAT TO DO.

The system prompt fixes who the model is and how it must behave; the user prompt is the actual request. One window, two very different jobs.

what the model keeps(Short-term and Long-term memory)

IT STOPPED FORGETTING.

Long-term memory carries facts, preferences, and past turns back into the window — so the model doesn't start every call from nothing.

what the model retrieves(RAG)

FETCH IT, DON'T TRAIN IT.

Retrieval-augmented generation pulls the relevant documents in at query time — knowledge the weights never had to memorize.

what the model can reach

IT CAN ACT, NOT JUST ANSWER.

Tools let the model call code, search, or an API and read the result back — turning a text generator into something that can act.

56.1%

Bury the answer in the middle of a long context and the model scores worse than with no documents at all.

88.3% when given only the gold document. "Lost in the Middle," Liu et al., TACL 2024.

so the job changed name

curate. compress. order.

Context engineering — "curating and maintaining the optimal set of tokens during inference." Named June 2025, defined by Anthropic that September.

03

HARNES

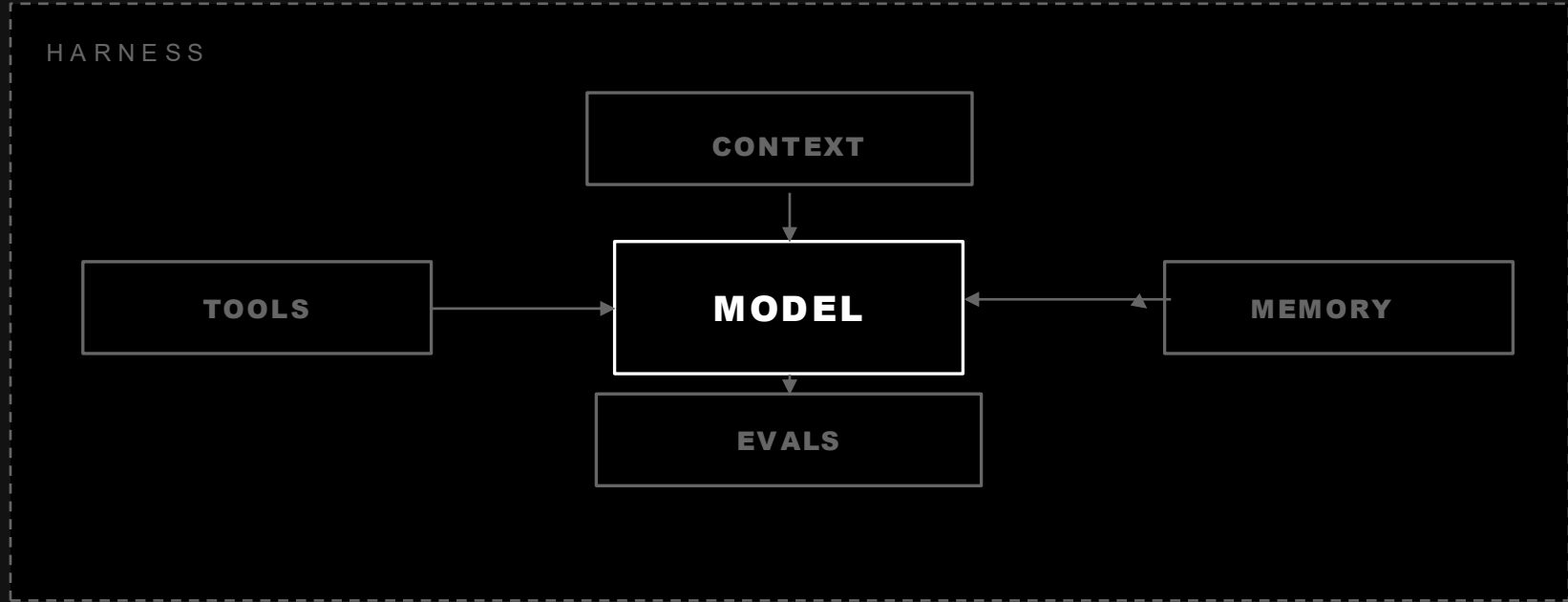
what surrounds the model?

an agent harness is

the execution environment.

It decides what context the model sees, what memory it keeps, what tools it can reach, and whether the output was any good. Everything outside the weights.

The model stopped being the product



Curate the input, hold the state, expose the tools, judge the output.

One lives inside the other.

CONTEXT ENGINEERING

**What goes
in the
window.**

Curate, compress, and order the tokens — before the model answers.

THE HARNESS

**Everything
around the
weights.**

Context, memory, tools, evals — and the loop that runs them.

Context engineering is one job the harness does — not a competing layer.

04

LOOPS

Model runs repeatedly until it completes the task

WHAT STARTED THE HYPE



Peter Steinberger



@steipete



Here's your monthly reminder that you shouldn't be prompting coding agents anymore.

You should be designing loops that prompt your agents.

11:58 AM · Jun 7, 2026 · **1.2M** Views



803



906



9.6K



6.9K



WHAT STARTED THE HYPE



Rohan Paul 
@rohanpaul_ai

Subscribe

"I don't prompt Claude anymore. I have loops running that prompt Claude and figuring out what to do. My job is to write loops. And this is transition we're going to see for the rest of the year."

– Boris Cherny, head of Claude Code at Anthropic.

THE DEFINITION

A repeating feedback loop that allows an AI to plan, act, evaluate, and improve its own work until it reaches a desired outcome.

Anthropic, "Building effective agents," December 2024. The whole idea, in five words.

Not one loop. Five shapes of loop.

01 **CORE AGENT**

Plan, act, evaluate, repeat until the task is done.

02 **VERIFICATION**

A second pass that checks the first: tests, critics, review.

03 **EVENT-DRIVEN**

No fixed start. It wakes on a webhook, a commit, a message.

04 **OPTIMIZATION**

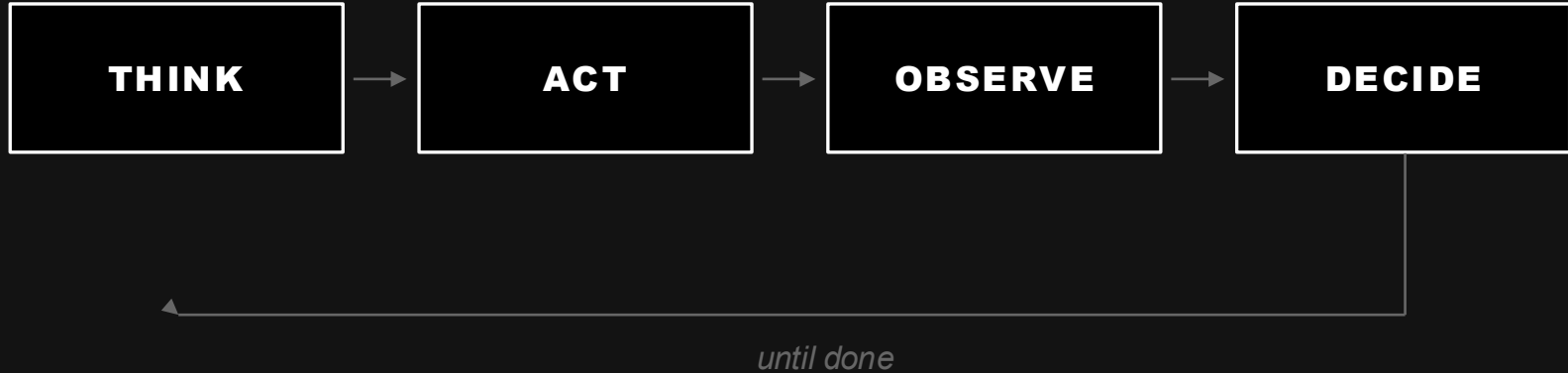
Hold the goal fixed, vary the attempt, keep the best one.

05 **SELF-IMPROVEMENT**

The loop edits the agent itself: prompts, tools, memory.

Same primitive. Different trigger, different stopping rule.

One pass became many.



The model no longer answers. It works.

2x

Since 2024, the task length an agent can finish has doubled roughly every 88 days.

At 50% reliability. METR, Time Horizon 1.1, January 2026.

05

GRAPH

or is a graph just a loop that branches?

WHAT STARTED THE HYPE



Peter Steinberger 🦀 🌐 📺

@steipete



Are we still talking loops or did we shift to graphs yet?

7:34 PM · Jul 17, 2026 · **3M** Views

💬 1.2K

↻ 1K

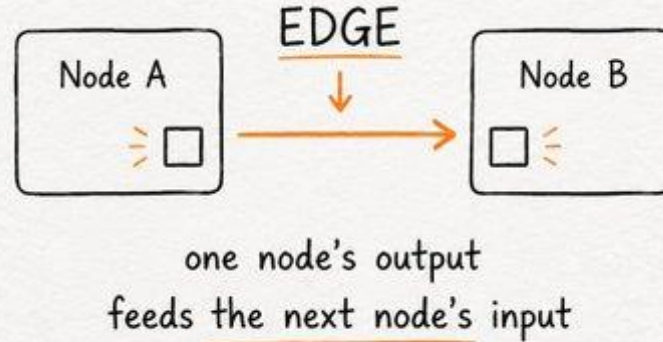
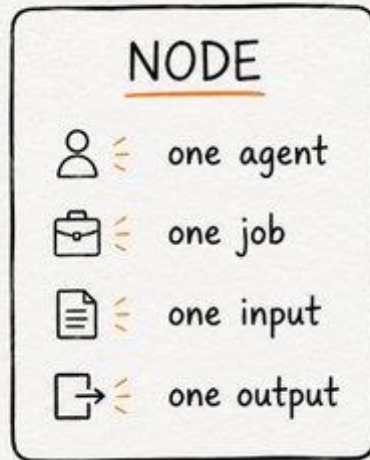
❤️ 7.7K

🔖 2.7K



Definition of Nodes & Edges

≡ Nodes & Edges ≡



One word, two definitions.

GRAPH AS DATA

**Entities
and edges
as the index.**

GraphRAG, Microsoft Research, open-sourced 2024.

GRAPH AS CONTROL

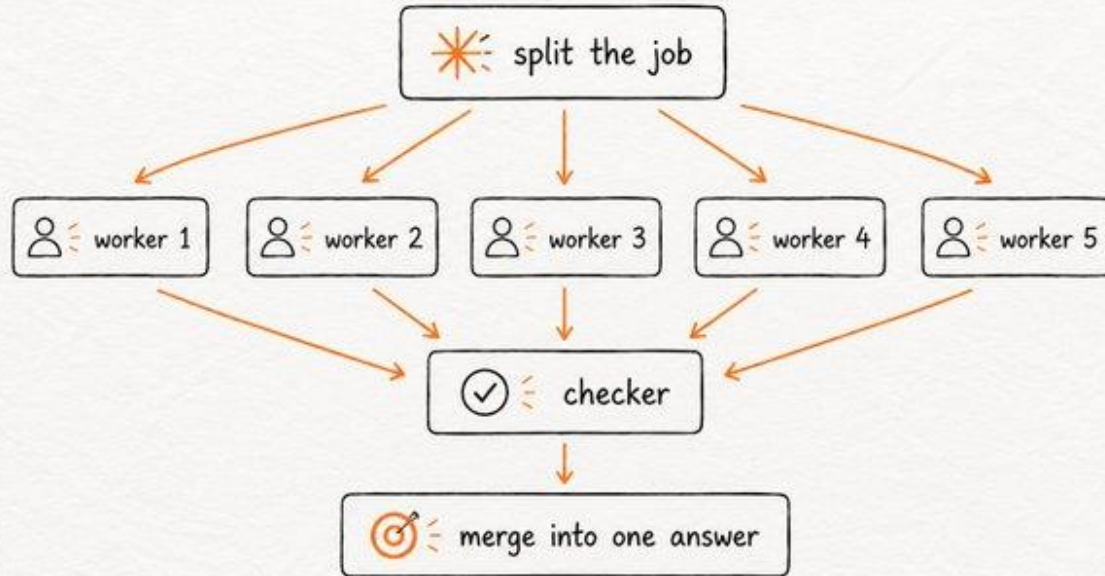
**Nodes
and edges
as the program.**

LangGraph, LangChain. 1.0 GA in October 2025.

If you say "graph" without saying which one, nobody knows what you built.

EXAMPLE OF GRAPH ENGINEERING

THE DIAMOND: fan out → reduce → synthesize.

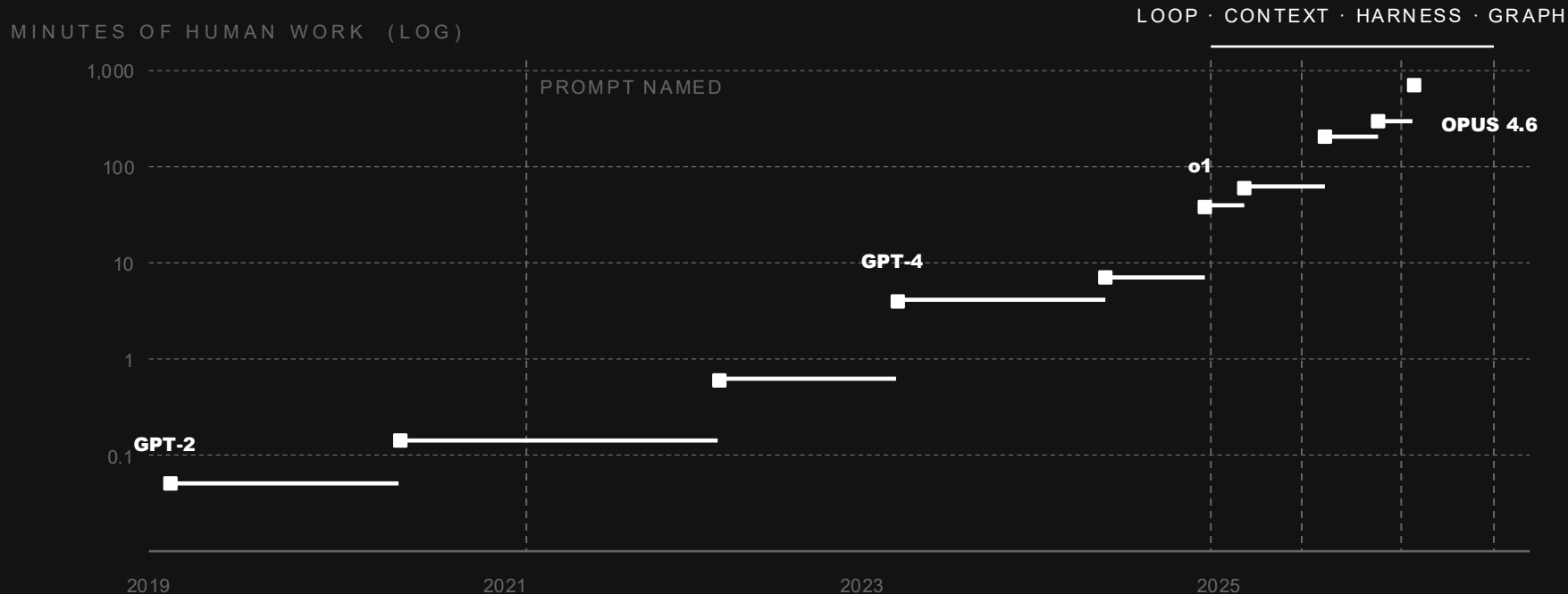


and the reason isn't control flow

one window becomes four.

Each loop works in its own context and hands back a summary. The orchestrator never sees the mess — which is a context problem solved with a shape.

The longer a model could work alone, the more we had to build around it.



3 seconds of work in 2019. Twelve hours in 2026 — and every layer arrived after the jump, not before.

METR 50% time horizon, TH1.1, data to May 2026. METR calls its own measurements above 16 hr unreliable.

An agent that reviews pull requests.

CONTEXT

The diff, the failing test output, and the three files it touches. Not the repo.

HARNESS

Sandboxed checkout, read-only git, a test runner it can call, a rubric it's graded on.

LOOP

Read, run tests, form a finding, check it against the code, revise. Stop when nothing new.

GRAPH

Three reviewers fan out — correctness, security, style. One node joins and dedupes.

Take away any one layer and it stops working.



FROM PROMPT TO GRAPH, WHATS NEXT?